

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 13
Simulation of TCP Congestion Control Algorithms

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Aim:

Simulation of Congestion Control (Reno and Tahoe) variants in NS2

```
tahoe.tcl
# Create a simulator object
set ns [new Simulator]
# Define different colors
# for data flows (for NAM)
$ns color 1 Blue
$ns color 2 Red
#Open the NAM trace file
set nf [open taho.nam w]
$ns namtrace-all $nf
set tf [open taho.tr w]
$ns trace-all $tf
#Define a 'finish' procedure
proc finish {} {
    global ns nf tf
    $ns flush-trace
    #Close the NAM trace file
    close $nf
    close $tf
    #Execute NAM on the trace file
    exec nam taho.nam &
    exec awk -f Throughput.awk taho.tr &
    exit 0 }
```

```
# Create four nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
```

```
# Create links between the nodes
```

```
$ns duplex-link $n0 $n1 10Mb 10ms DropTail
$ns duplex-link $n1 $n2 2Mb 10ms DropTail
```

```
# Set Queue Size of link (n2-n3) to 10
$ns queue-limit $n0 $n1 10
```

```
# Give node position (for NAM)
#$ns duplex-link-op $n0 $n2 orient right
$ns duplex-link-op $n0 $n1 orient right-down
$ns duplex-link-op $n2 $n1 orient left-down
```

```
# Monitor the queue for link (n2-n3). (for NAM)
$ns duplex-link-op $n0 $n1 queuePos 0.5
```

```
# Setup a TCP connection
set tcp [new Agent/TCP]
$tcp set window_ 10 ;# Set the window size (e.g., 10 packets)
$tcp set packetSize_ 1000 ;# Set the packet size (e.g., 1000
bytes)
$tcp set timeout_ 1.0 ;# Set the timeout (e.g., 1.0 seconds)
$ns attach-agent $n0 $tcp
```

```
set sink [new Agent/TCPSink]
$ns attach-agent $n2 $sink
$ns connect $tcp $sink
$tcp set fid_ 1
```

```
# Setup a FTP over TCP connection
```

```
set ftp [new Application/FTP]
$ftp attach-agent $tcp
$ftp set type_ FTP
```

```
# Schedule events for the CBR and FTP agents
#$ns at 0.1 "$cbr start"
$ns at 0.1 "$ftp start"
$ns at 4.0 "$ftp stop"
```

```
#$ns at 4.5 "$cbr stop"
```

```
# Detach tcp and sink agents
# (not really necessary)
#$ns at 4.5 "$ns detach-agent $n0 $tcp ; $ns detach-agent $n2
$sink"
```

```
# Call the finish procedure after
# 5 seconds of simulation time
$ns at 5.0 "finish"
```

```
# procedure to plot the congestion window
proc plotWindow {tcpSource outfile} {
```

```
    global ns
    set now [$ns now]
    set cwnd [$tcpSource set cwnd_]
```

```
# the data is recorded in a file called congestion.xg (this can
be plotted # using xgraph or gnuplot. this example uses xgraph
to plot the cwnd_
```

```
    puts $outfile "$now $cwnd"
```

```
    $ns at [expr $now+0.1] "plotWindow $tcpSource $outfile"
}
```

```
set outfile [open "taho.xg" w]
$ns at 0.0 "plotWindow $tcp $outfile"
```

```
# Run the simulation
$ns run
```

reno.tcl

```
#Create a simulator object
```

```
set ns [new Simulator]

#Open the nam file basic1.nam and the variable-trace file
basic1.tr
set namfile [open basic1.nam w]
$ns namtrace-all $namfile

set tracefile [open basic1.tr w]
$ns trace-all $tracefile

#Define a 'finish' procedure
proc finish {} {

    global ns namfile tracefile
    $ns flush-trace
    close $namfile
    close $tracefile

    exit 0 }

#Create the network nodes
set A [$ns node]
set R [$ns node]
set B [$ns node]

#Create a duplex link between the nodes

$ns duplex-link $A $R 10Mb 10ms DropTail
$ns duplex-link $R $B 800Kb 50ms DropTail

# The queue size at $R is to be 7, including the packet being
sent

$ns queue-limit $R $B 7

# some hints for nam
```

```
# color packets of flow 0 red
$ns color 0 Red
$ns duplex-link-op $A $R orient right
$ns duplex-link-op $R $B orient right
$ns duplex-link-op $R $B queuePos 0.5
```

```
# Create a TCP sending agent and attach it to A
set tcp0 [new Agent/TCP/Reno]
# We make our one-and-only flow be flow 0
$tcp0 set class_ 0
```

```
$tcp0 set window_ 100
$tcp0 set packetSize_ 960
$ns attach-agent $A $tcp0
```

```
# Let's trace some variables
$tcp0 attach $tracefile
$tcp0 tracevar cwnd_
$tcp0 tracevar ssthresh_
$tcp0 tracevar ack_
```

```
$tcp0 tracevar maxseq_
```

```
#Create a TCP receive agent (a traffic sink) and attach it to B
set end0 [new Agent/TCPSink]
$ns attach-agent $B $end0
```

```
#Connect the traffic source with the traffic sink
$ns connect $tcp0 $end0
```

```
#Schedule the connection data flow; start sending data at T=0,
stop at T=10.0
set myftp [new Application/FTP]
$myftp attach-agent $tcp0
```

```
$ns at 0.0 "$myftp start"
$ns at 10.0 "finish"
```

```
proc plotWindow {tcpSource outfile} {  
    global ns  
  
    set now [$ns now]  
    set cwnd [$tcpSource set cwnd_  
  
    # the data is recorded in a file called congestion.xg (this can  
    be plotted # using xgraph or gnuplot. this example uses xgraph  
    to plot the cwnd_  
  
    puts $outfile "$now $cwnd"  
  
    $ns at [expr $now+0.1] "plotWindow $tcpSource $outfile"  
}  
  
set outfile [open "reno.xg" w]  
$ns at 0.0 "plotWindow $tcp0 $outfile"  
  
#Run the simulation  
$ns run
```

